Assignment 02

1.1 各国地震死亡人数排名

缺失值处理：sum(min_count=1) 至少有一个非缺失值才计算总和，全部缺失返回 NaN

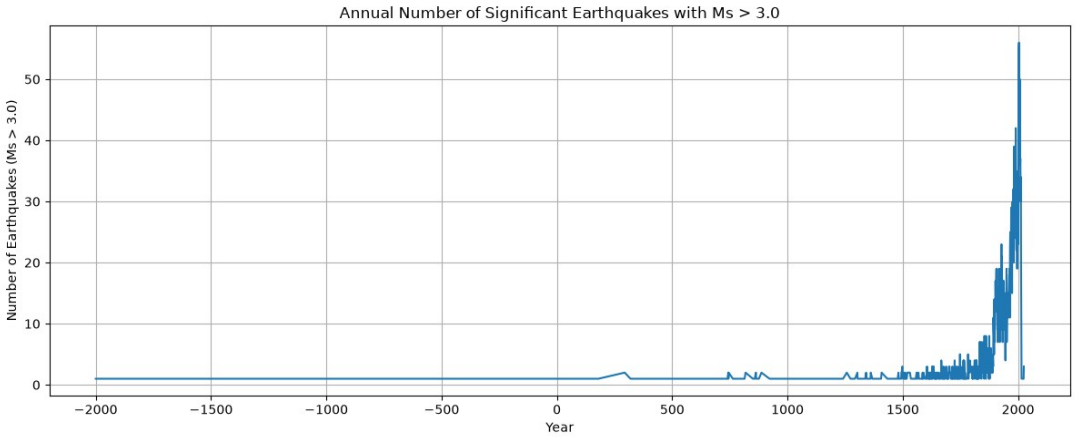
结果如下：

```
Out[54]: Country
         CHINA           2106526.0
         TURKEY          1008867.0
         IRAN             761656.0
         SYRIA            488326.0
         ITALY            417297.0
         HAITI            323782.0
         JAPAN            319510.0
         AZERBAIJAN       319251.0
         INDONESIA        283485.0
         ARMENIA          175000.0
         PAKISTAN         143714.0
         ECUADOR          134445.0
         TURKMENISTAN     110412.0
         PERU              97009.0
         GREECE            83658.0
         PORTUGAL          82572.0
         IRAQ              70200.0
         CHILE             70175.0
         INDIA             63413.0
         ISRAEL            58368.0
         Name: Total Deaths, dtype: float64
```

1.2 全球每年震级大于 3.0 的地震数量

删除震级或年份缺失的记录后，存在 511 组有效数据。

震级大于 3.0 地震数量有增加的趋势，1900 年之后数量大幅度增加。可能是因为地震监测技术的进步，导致更多的地震被检测到并记录下来。



1.3

删除 Ms 缺失的数据

Country	Total_Earthquakes	Largest_Earthquake_Date	Largest_Earthquake_Location
CHINA	626	1920.0-12.0-16.0	CHINA: GANSU PROVINCE, SHANXI PROVINCE
JAPAN	429	869.0-7.0-13.0	JAPAN: SANRIKU
INDONESIA	427	2004.0-12.0-26.0	INDONESIA: SUMATRA: ACEH: OFF WEST COAST
IRAN	385	856.0-12.0-22.0	IRAN: DAMGHAN, QUMIS
TURKEY	359	1939.0-12.0-26.0	TURKEY: ERZINCAN
ITALY	338	1915.0-1.0-13.0	ITALY: MARSICA, AVEZZANO, ABRUZZI
GREECE	296	365.0-7.0-21.0	GREECE: CRETE: KNOSSOS
USA	283	1957.0-3.0-9.0	ALASKA
PHILIPPINES	233	1897.0-9.0-21.0	PHILIPPINES: MINDANAO, ZAMBOANGA, SULU, ISABELA
MEXICO	217	1787.0-3.0-28.0	MEXICO: SAN MARCOS, OAXACA
CHILE	200	1730.0-7.0-8.0	CHILE: VALPARAISO
PERU	195	1619.0-2.0-14.0	PERU: TRUJILLO, PIURA, SANTA
RUSSIA	159	1952.0-11.0-4.0	RUSSIA: KAMCHATKA PENINSULA
PAPUA NEW GUINEA	107	2000.0-11.0-16.0	PAPUA NEW GUINEA: NEW IRELAND, DUKE OF YORK
TAIWAN	103	1920.0-6.0-5.0	TAIWAN
INDIA	102	1897.0-6.0-12.0	INDIA: ASSAM; BANGLADESH
COLOMBIA	83	1904.0-1.0-20.0	PANAMA-COLOMBIA
AFGHANISTAN	72	1909.0-7.0-7.0	AFGHANISTAN: HINDU-KUSH
NEW ZEALAND	72	1826.0-nan-nan	NEW ZEALAND: FJORDLAND
ECUADOR	70	1906.0-1.0-31.0	ECUADOR: OFF COAST

2.

1) 读取 **DATE** 和 **TMP** 两列数据

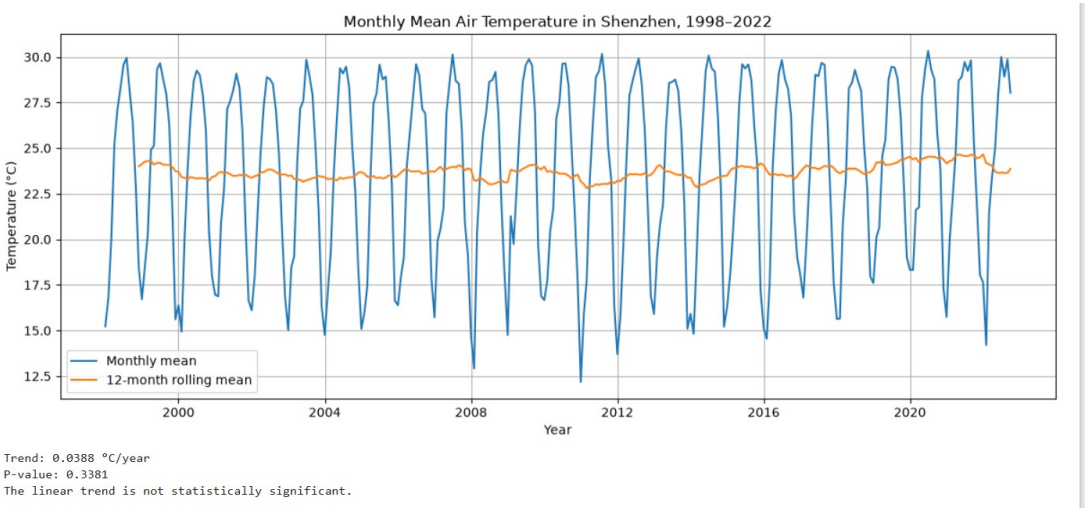
2) 将 **DATE** 转换为日期时间格式，无法转换的记录设为缺失

3) 将气温数值除以 **10**，转换为摄氏度

4) 将**+9999** 这一缺失标记转换为缺失值

5) 保留有效日期、有效气温及质量代码为 **1**（通过全部质量控制检查）或 **5**（通过全部质量控制检查，数据来自 **NCEI**）的记录

6) 按月份分组并计算算术平均值，没有有效观测的月份，保留缺失值



深圳宝安国际机场 1998—2022 年月平均气温总体趋于稳定

3.

3.1

SID	NAME	WMO_WIND
2015293N13266	PATRICIA	185.0
1980214N11330	ALLEN	165.0
2005289N18282	WILMA	160.0
1997253N12255	LINDA	160.0
1988253N12306	GILBERT	160.0
1935241N23291	NaN	160.0
2019236N10314	DORIAN	160.0
2009288N07267	RICK	155.0
1998295N12284	MITCH	155.0
2017242N16333	IRMA	155.0

3.2

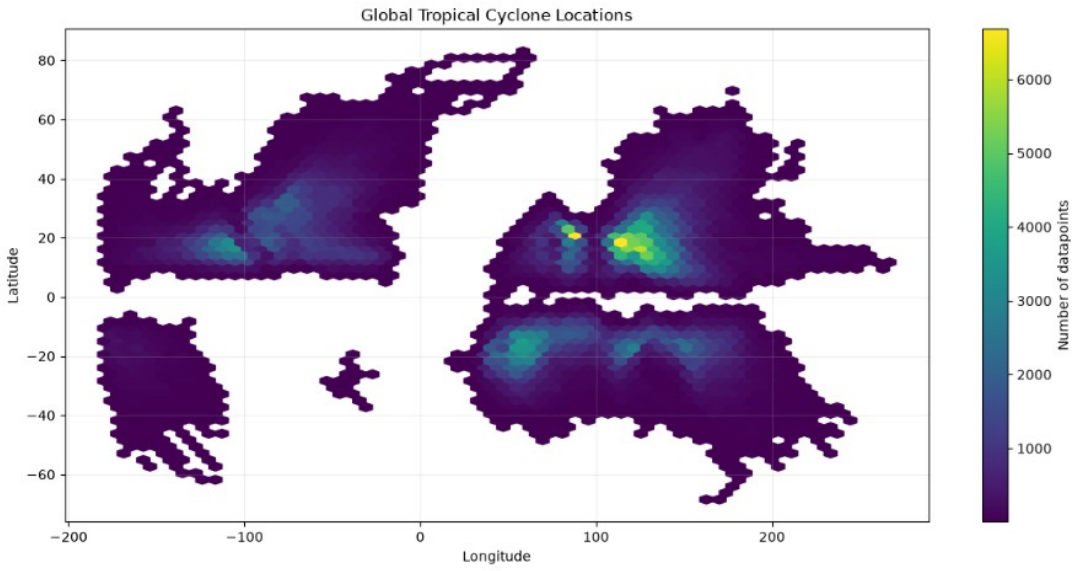
SID	NAME	WMO_WIND
2015293N13266	PATRICIA	185.0
1980214N11330	ALLEN	165.0
2005289N18282	WILMA	160.0
1997253N12255	LINDA	160.0
1988253N12306	GILBERT	160.0
1935241N23291	NaN	160.0
2019236N10314	DORIAN	160.0
2009288N07267	RICK	155.0
1998295N12284	MITCH	155.0
2017242N16333	IRMA	155.0
2005261N21290	RITA	155.0
2017260N12310	MARIA	150.0
1955265N13307	JANET	150.0
1979238N12324	DAVID	150.0
1977242N27272	ANITA	150.0
2007244N12303	FELIX	150.0
2007225N12331	DEAN	150.0
2005236N23285	KATRINA	150.0
1994222N11267	JOHN	150.0
1992230N11325	ANDREW	150.0

3.3

```
BASIN
WP    236576
SI    160668
SP     67119
EP     62412
NI     55401
SA       119
Name: count, dtype: int64
```

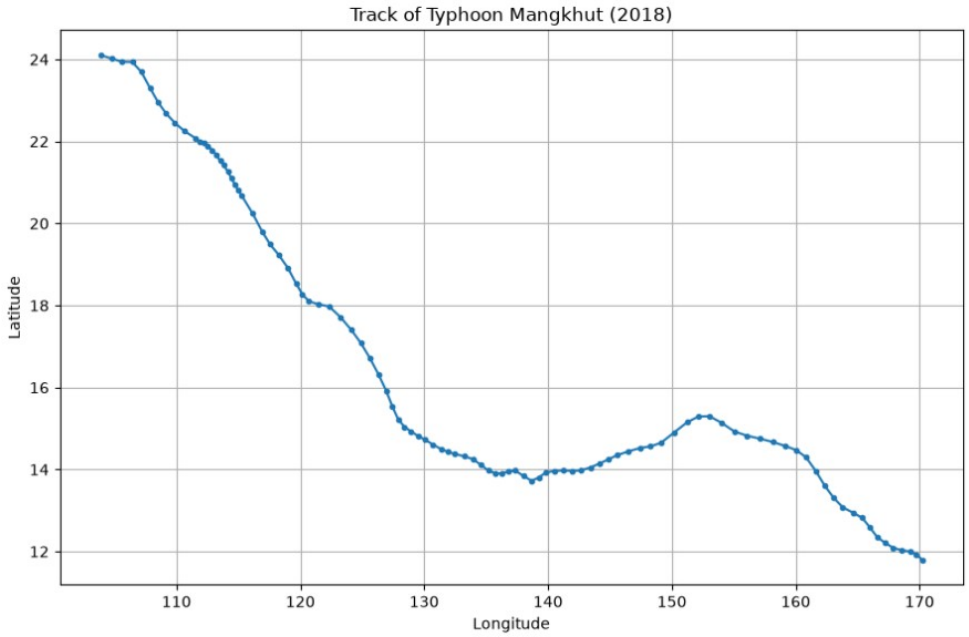
3.4

删除 LAT 或 LON 缺失的记录



3.5

筛选 NAME = MANGKHUT 且 ISO_TIME 年份为 2018 的记录，删除缺失经纬度的点，并按观测时间排序，随后绘制经度为横轴、纬度为纵轴的轨迹线。



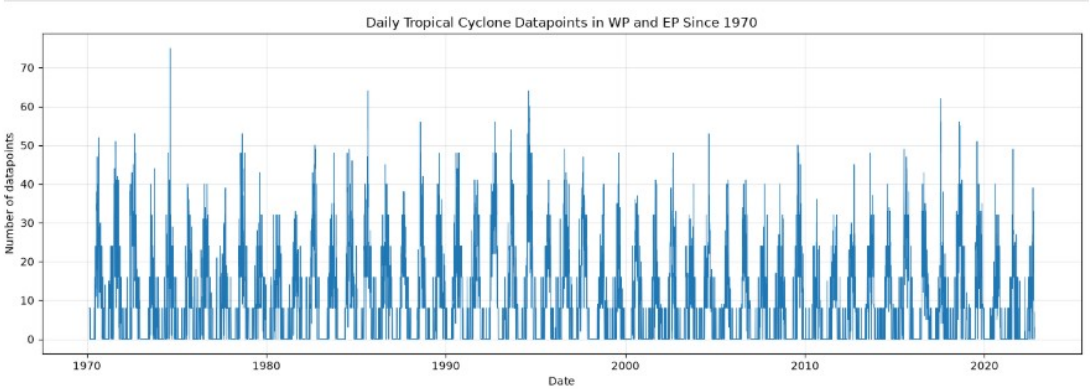
3.6

```
                SID  SEASON  NUMBER BASIN SUBBASIN   NAME  \
350393  1970050N07151    1970      22    WP       MM  NANCY
350394  1970050N07151    1970      22    WP       MM  NANCY
350395  1970050N07151    1970      22    WP       MM  NANCY
350396  1970050N07151    1970      22    WP       MM  NANCY
350397  1970050N07151    1970      22    WP       MM  NANCY

                ISO_TIME NATURE      LAT      LON  WMO_WIND WMO_PRES  \
350393 1970-02-19 00:00:00     TS  7.00000  151.400       NaN     1006
350394 1970-02-19 03:00:00     TS  7.24752  151.205       NaN
350395 1970-02-19 06:00:00     TS  7.50000  151.000       NaN     1002
350396 1970-02-19 09:00:00     TS  7.75747  150.772       NaN
350397 1970-02-19 12:00:00     TS  8.00000  150.500       NaN      998

       WMO_AGENCY TRACK_TYPE  DIST2LAND LANDFALL           IFLAG
350393      tokyo       main       1088     1088  _OO____________
350394                  main       1109     1109  _PP____________
350395      tokyo       main       1143     1143  _OO____________
350396                  main       1171     1168  _PP____________
350397      tokyo       main       1182     1179  _OO____________
Number of datapoints: 176352
Basins: <ArrowStringArray>
['WP', 'EP']
Length: 2, dtype: str
```

3.7

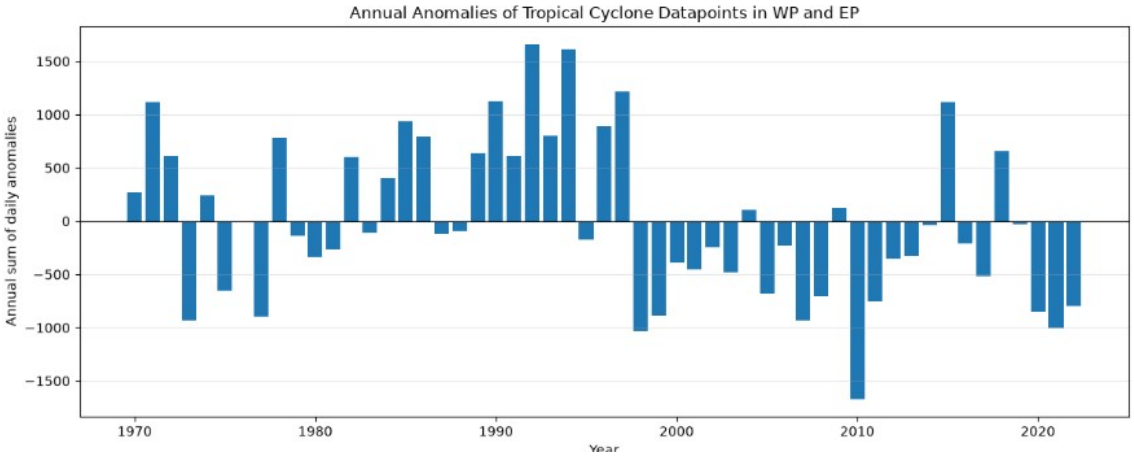


3.8

```
Date
1    1.596154
2    1.384615
3    1.423077
4    1.788462
5    2.019231
dtype: float64
```

3.9

正值代表全年观测点数量累计高于每日气候态之和，负值代表低于该基准。1992、1994、1997、1990、2015、20010、1998、2021、2007、1973 的飓风活动突出



```
Years with the highest annual anomalies:
Date
1992-12-31    1660.725327
1994-12-31    1611.725327
1997-12-31    1215.725327
1990-12-31    1121.725327
2015-12-31    1119.725327
dtype: float64

Years with the lowest annual anomalies:
Date
2010-12-31   -1668.274673
1998-12-31   -1033.274673
2021-12-31   -1004.274673
2007-12-31    -935.274673
1973-12-31    -933.274673
dtype: float64
```

4.2025 年 4-10 月伊利湖西部表层水温分析

1）数据来源：NOAA-GLERL WLE Weekly Data

2025 年数据链接：

https://apps.glerl.noaa.gov/proxy/proxy-husky.php?dir=%22habs-and-hypoxia%22&file=%222025_WLE_Weekly_Datashare_CSV.csv%22

将 Date 列转换为日期格式，将 Temp_C 列转换为数值格式。无法转换的记录设为缺失值。根据 Sample_Depth_category 列筛选类别为 S 的表层水温数据。统计表层水样中的缺失温度值和缺失日期值，并计算缺失温度所占的百分比并删除日期或温度缺失的记录。

2）统计分析

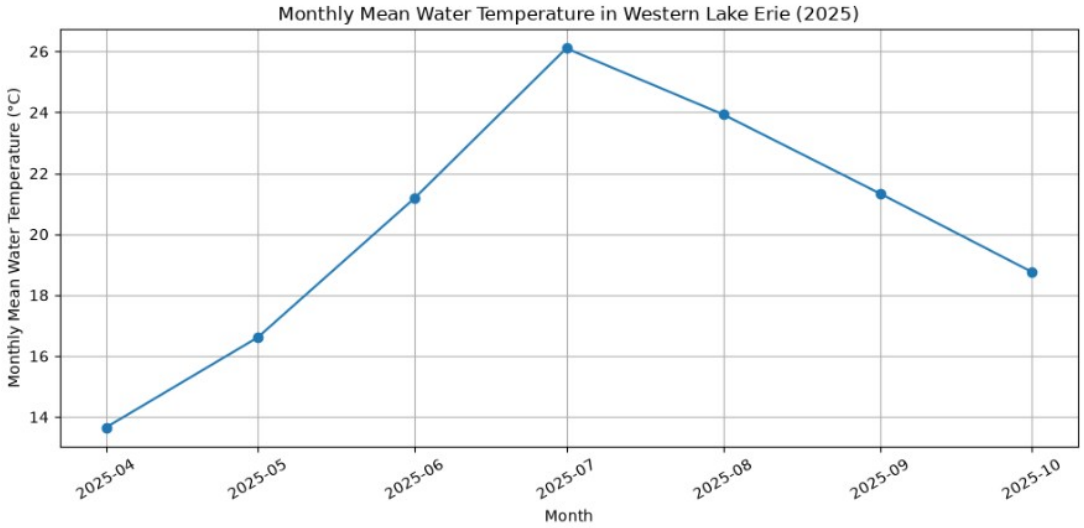
共获得 155 条有效表层水温观测值

```
Descriptive Statistics
count    155.00
mean      21.65
std        3.88
min       11.43
25%       18.66
50%       21.85
75%       25.15
max       28.20
Name: Temp_C, dtype: float64
Range: 16.76 °C
```

3）计算月平均水温变化

```
Monthly Temperature Results
            Mean_Temp_C  Sample_Count  Monthly_Change_C
Date
2025-04-01        13.66             8               NaN
2025-05-01        16.61             8              2.95
2025-06-01        21.18            32              4.57
2025-07-01        26.10            32              4.92
2025-08-01        23.93            18             -2.17
2025-09-01        21.33            39             -2.59
2025-10-01        18.76            18             -2.57
Monthly Temperature Changes
2025-04 to 2025-07: 12.44 °C
2025-07 to 2025-10: -7.34 °C
```

4）月平均水温时间序列图



a. 春季至初夏阶段（4 月至 6 月）： 月平均水温从 13.66 °C 上升至 21.18 °C，呈现持续升温的特征。

b. 夏季阶段（6 月至 7 月）： 水温继续上升，并在 7 月达到研究期间的最高月平均值 26.10 °C。

c. 夏末至秋季阶段（8 月至 10 月）： 水温由 23.93 °C 逐渐下降至 18.76 °C，表现出持续降温的特征。

伊利湖西部表层水温呈明显季节变化趋势。